

Q6. Bigram Probabilities and the Zero-Probability Problem

You are given the following bigram counts from a small training corpus:

Previous word	Next words (with counts)
<s>	I: 2 , deep: 1
I	love: 2
love	NLP: 1 , deep: 1
deep	learning: 2
learning	</s>: 1 , is: 1
NLP	</s>: 1
is	fun: 1
fun	</s>: 1
ate	lunch: 6 , dinner: 3 , a: 2 , the: 1

Tasks:

1. Bigram Sentence Probabilities

Using maximum likelihood estimation (MLE):

$$P(I|<s>) = 2/3$$

$$P(\text{love}|I) = 2/2 = 1$$

$$P(\text{NLP}|\text{love}) = 1/2,$$

$$P(\text{deep}|\text{love}) = 1/2$$

$$P(</s>|\text{NLP}) = 1/1 = 1$$

$$P(\text{learning}|\text{deep}) = 2/2 = 1$$

$$P(</s>|\text{learning}) = 1/2$$

$$P(w_i | w_{i-1}) = \frac{C(w_{i-1}, w_i)}{C(w_{i-1})}$$

- Compute the probability of sentence S1: <s> I love NLP </s>.

$$P(S1) = 2/3 * 1 * 1/2 * 1 = 0.3333$$

- Compute the probability of sentence S2: <s> I love deep learning </s>.

$$P(S2) = 2/3 * 1 * 1/2 * 1 * 1/2 = 0.1667$$

- Which sentence is more probable under the bigram model?

More Probable is S1

2. Zero-Probability Problem

Using the same table, compute:

- $P(\text{noodle}|\text{ate})$ with MLE.

$$P(\text{noodle}|\text{ate}) = 0/12 = 0$$

3. Explain why this probability creates problems when computing sentence probabilities or perplexity.

Any sentence containing the unseen bigram gets total probability 0, log-probability $-\infty$, and **perplexity becomes infinite**, breaking evaluation and ranking.

4. Apply Laplace smoothing (Add-1) to recompute $P(\text{noodle}|\text{ate})$. Assume the vocabulary size is 10 and total count after “ate” is 12.

Add-1 (Laplace) smoothing ($V=10$):

$$P(\text{noodle}|\text{ate}) = (0+1)/(12+10) = 0.0455$$